

TABLE OF CONTENTS

TOPIC NUMBER	TOPIC NAMES	PAGE NO
Chapter-1		
Alcohol Detector System with Motor Locking Using Arduino Uno		
1.1	Abstract	7
1.2	Introduction	7
1.3	Objectives	7-8
1.4	Literature survey	8-9
1.5	Problem statement	9-10
1.6	Methodology	10
1.7	Flow chart	11
1.8	Block diagram	12
1.9	Circuit diagram	13
1.10	Components used	13-30
1.11	Procedure for upload a code	31
1.12	Source code	31-32
1.13	Result	32-33
1.14	Applications	33-34
1.15	Advantages	34-35
1.16	Disadvantages	35
1.17	Future scope	35
1.18	Video link	35
Chapter-2		
Auto Electric Vehicles		
2.1	Introduction	36
2.2	Tata Nexon Ev	37-38
2.3	Battery charger Modular	38-39
2.4	Godawari Electric scooter	39-40

2.5	LIRON LIB Power	40
2.6	I-3 Sensopart	41
2.7	Hellermann Tyton	42-43
Chapter-3 Basics of Solar		
3.1	Learning Outcomes	44
3.2	How Does Electricity Reach Our Homes?	44-45
3.3	Renewable Energy Sources	46-47
3.4	Solar Energy	47-48
3.5	Basic Terminologies	49
3.6	Photovoltaic Effect	50
3.7	Solar Power Plant Capacity	50
3.8	Net Metering Arrangement	52
3.9	Solar Panels / Modules	53
3.10	Module Mounting Structure (Mms)	54
3.11	Inverter	55
3.12	Reversible Reductions in Output Power	55
3.13	Overview Of Fenice Energy Products	56
3.14	Subsidy And Empanelment	56
3.15	Difference in Module	57
3.16	Photos	58-59
Chapter-4 My Entrepreneur Journey		
4.1	Introduction	60-61
4.2	Objectives	61
4.3	Outcomes	62
Chapter-5		
5.1	Conclusion	63-34
5.2	References	64

LISTS OF FIGURES

FIGURE NUMBER	FIGURE NAME	PAGE NO
Chapter-1		
Alcohol Detector System with Motor Locking using Arduino Uno		
1.1	Flow chart	11
1.2	Block diagram	12
1.3	Circuit diagram	13
1.4	Arduino UNO	14
1.5	MQ-3 alcohol sensor	16
1.6	Buzzer	18
1.7	LED	18
1.8	5v DC motor	20
1.9	Relay module schematics	22
1.10	5v Relay	24
1.11	Resistor	25
1.12	Male to Male jumper wires	26
1.13	Male to Female jumper wires	27
1.14	Push button	28
1.15	Push button pinout/connections in released state	29
1.16	Push button pinout/connections in pressed state	29
1.17	System is off state	32
1.18	System ON state without Alcoholic data	32
1.19	System ON state with Alcoholic data	33
Chapter-2		
Auto Electric Vehicles		

2.1	Entrance	36
2.2	Tata Nexon EV	37
2.3	Battery charger modular	38
2.4	Godawari electric scooter	39
2.5	Lithium ion batteries	40
2.6	Vision sensors	41
2.7	Cables	42
2.8	Group photo	43
Chapter-3 Basics of Solar		
3.1	Solar Energy system	44
3.2	Single line diagram of Power system	44
3.3	Power Plant	45
3.4	Renewable energy sources	46
3.5	Solar Energy	47
3.6	Solar Thermal	48
3.7	Solar PV	48
3.8	Basic Terminologies	49
3.9	Photovoltaic Effect	50
3.10	Power Estimation	50
3.11	Solar Power System	52
3.12	Net Meter	52
3.13	Solar Panel System	53
3.14	Module Mounting Structure	54
3.15	Inverter	55
3.16	Reversible Reductions in Output Power	55
3.17	Overview of Fenice Energy products	56
3.18	Photo 1	58

3.19	Photo 2	58
3.20	Photo 3	59
3.21	Photo 4	59
Chapter-4 My Entrepreneur Journey		
4.1	Photo 1	58
4.2	Photo 2	58
4.3	Photo 3	59
4.4	Photo 4	59

LISTS OF TABLES

TABLE NUMBER	TABLE NAME	PAGE NO
1.1	MQ-3 Sensor Standard work condition	116
1.2	MQ-3 Sensor Environment condition	17
1.3	MQ-3 Sensor Sensitivity characteristic condition	17
1.4	Push Button Specifications	30
3.1	Types of crystalline modules	53
3.2	Difference in module	57

CHAPTER-1

ALCOHOL DETECTOR SYSTEM WITH MOTOR LOCKING USING ARDUINO UNO

1.1 ABSTRACT

Nowadays there are many accidents occur due to drivers consuming alcohol while operating vehicles. Drunk driving remains a leading cause of accidents in numerous countries and it is a major cause of accidents in worldwide. The Alcohol Detection in Car project aims to enhance the safety of vehicle occupants by implementing an automatic motor locking system and this project utilizes an alcohol sensor to detect alcoholic breath. Upon detecting alcohol, an intelligent ATmega328P microcontroller activates a process to lock the Motor using a DC motor. By proactively addressing drunk driving, this project enhances safety for passengers. If the alcohol level exceeds the set limit, the system locks the Motor and activates a buzzer and LED, thereby preventing accidents.

Keywords: Arduino UNO, MQ3 Alcohol Sensor, DC motor, Relay, Buzzer, LED.

1.2 INTRODUCTION

We hear a lot about accidents caused by drunk driving, which not only jeopardizes the stability of the driver but also poses a severe inconvenience to other road users, often resulting in fatalities. Our project focuses on preventing drunk driving by utilizing a compact Arduino UNO board and programming in embedded C. We employ an MQ3 sensor, a DC motor, an Arduino board, and a Relay for this purpose. The Arduino is responsible for programming and interfacing, while the Relay module facilitates the switching of high voltage or high current devices using a 5V signal from the Arduino. The MQ3 sensor, positioned on the steering of the car, detects alcohol molecules. When a driver attempts to operate the vehicle under the influence, the MQ3 sensor detects the presence of alcohol. If the alcohol concentration exceeds the predetermined threshold of 40%, a signal is sent to the Arduino, which subsequently halts the operation of the motor.

1.3 OBJECTIVES

1. To provide alcohol detection with Motor locking system which ensures the driver safety.
2. To avoid road accidents due to driver under alcohol influence.

1.4 LITERATURE SURVEY

1. The author has proposed a method to detect alcohol but uses GPS and GSM module which increases the overall cost which could be avoided. In our project, we are using a siren which will be more cost efficient. Use of siren will alert the people nearby and hence any kind of necessary action can be taken. [1]
2. The authors propose to use a smart helmet to avoid accidents. There are many flaws with their design. A major shortcoming is the limitation of the its application to only vehicles which use helmets, i.e. 2 wheelers which is not a feasible idea while driving, especially for short distances. Another drawback being, the system when implemented makes the helmet too heavy which is not favourable for driving. Also, they have used an expensive microcontroller whereas we are using open-source hardware, which is very cheap. [2]
3. The authors discuss about complex health monitoring systems and infrared sensor to detect the presence of alcohol. A major drawback of this system is the possibility of a false alarm. The system is designed in a manner that even a slight change in some particular condition can result in ringing false alarms even though everything was normal. In our project, we are using only the required technology thereby making the system more reliable and cost effective when implemented. [3]
4. The authors have proposed a system to prevent the accidents due to drunken driving. Major drawback of this system is that they have used PIC16F877A microcontroller which is not as useful as Arduino Uno microcontroller that we are using. Also, they have used an old design system which is not useful and increases the overall cost of the system which makes it expensive and somewhat unaffordable to certain segments of society thereby limiting its scope to be used. Hence, our system is more cost effective and can be easily afforded. [4]
5. The authors discuss the problem of drunken driving and propose to solve it by suggesting a system. However, the major drawback of their system is that they are using MQ2 alcohol sensor which is not accurate and is not specifically sensitive to alcohol. In our project, we are using a MQ3 sensor which is designed to be accurate towards alcohol detection which gives more accurate results and saves from raising false alarms. Also, they have used a PIC microcontroller which is expensive when compared to Arduino Uno which is open sourced. [5]

6. The author has tried to design a system to tackle the problem of helmet negligence and alcohol detection at the same time. The system is overly complex. Also, the author has used P89V51RD2 microcontroller, which is expensive when compared to Arduino Uno. Also, this system has a limited scope of usage because it can work only with 2 wheelers and not with any other segment of vehicle. However, our system can be integrated to not only 2 wheelers, but also with any kind of vehicle thereby preventing more accidents and saving more people. [6]

Outcome of Literature Survey:

Through the literature survey, we discovered that we can address the issue of road accidents caused by drunk driving and potentially save numerous lives by implementing this project in automobiles. By doing so, we aim to capitalize on its benefits and effectively reduce the occurrence of accidents.

1.5 PROBLEM STATEMENT

Design an Alcohol Detector System with Motor Locking using Arduino Uno to address the growing concern of alcohol-impaired driving, which poses a significant risk to road safety. The system aims to prevent individuals under the influence of alcohol from operating vehicles by integrating an alcohol sensor with an Arduino Uno microcontroller. The primary objective is to develop a reliable and cost-effective solution that detects alcohol levels in a person's breath and, upon detecting an unsafe level, triggers a motor locking mechanism in the vehicle.

The key components of the system include an alcohol sensor, Arduino Uno microcontroller, a motor control module, and a locking mechanism. The alcohol sensor will be responsible for accurately measuring alcohol concentration in the breath of the individual. The Arduino Uno will process this data and, based on a predefined threshold, activate the motor control module to engage the locking mechanism, preventing the vehicle from starting.

Additionally, the system should be designed to have user-friendly features, such as an LED indicator, to communicate the status of the alcohol level and whether the motor is locked or unlocked. The system should be robust, responsive, and easily integrable into various vehicle models.

In summary, the Alcohol Detector System with Motor Locking using Arduino Uno aims to provide a proactive solution to reduce instances of drunk driving, promoting road safety and

minimizing the associated risks. The system should be reliable, cost-effective, and userfriendly, contributing to the overall efforts to enhance public safety on the roads.

1.6 METHODOLOGY

The alcohol detection with Motor locking system helps to reduce accidents that occur due to drunk driving. The MQ-3 sensor detects the presence of alcohol in the surroundings. The sensor provides output based on the concentration; higher conductivity of the MQ-3 sensor indicates higher alcohol levels, which in turn sends readings to the Arduino UNO. If the reading exceeds the threshold level, the Arduino UNO will stop the DC motor. The LED will also blink if the distance is less than the safe distance, indicating to other vehicles. In this project, we focus on three main operations: sensing alcohol, alerting the driver, and motor locking. The complete system uses the Arduino UNO (Based on ATMEGA 328P), MQ3 sensor, buzzer, and motor drive, all handled by the Arduino UNO. Once the sensor detects alcohol, it transfers the information to the Arduino system, which prevents the motor from starting and locks it. Our proposed work consists of various units: the power supply unit, alcohol detection unit, motor locking unit, alarm unit, and indicating unit. A DC motor is used as the car engine to demonstrate the concept of motor/engine locking. The ATmega328, under the brand name of Arduino UNO, continuously monitors the output from the alcohol sensor. The Arduino UNO sketch, which provides the programming environment, is used to write the code.

Using the Arduino Uno microcontroller, we propose to design a system comprising an alcohol sensor, MQ3, to detect the presence of alcohol by analyzing a person's breath. It shuts down the motor when a specific amount of alcohol is detected, preventing mishaps or accidents caused by intoxicated drivers taking control of the vehicle. Hence, drunk driving is controlled, minimizing loss of life and property.

1.7 FLOW CHART

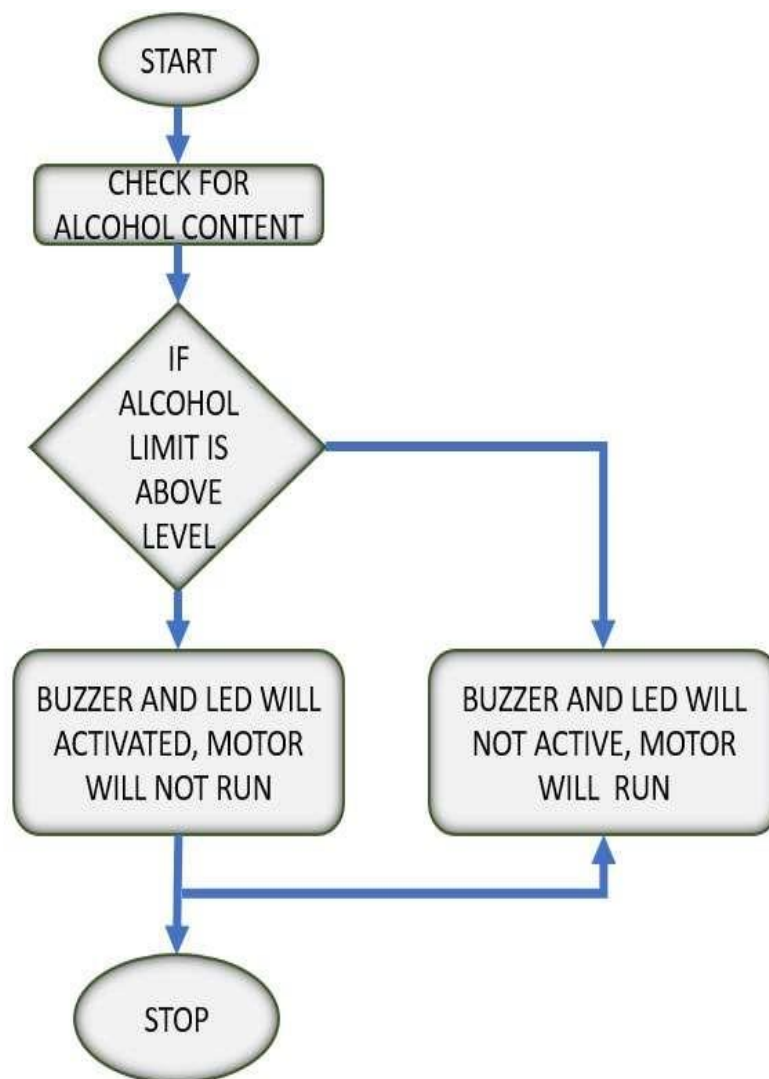


Fig.1.1: Flow chart

STEP 1: Start

STEP 2: Alcohol detection from the drunken drivers

STEP 3: Measuring the alcohol level of the driver STEP

3: If the Alcohol level above the limit.

STEP 4: Turn off the Motor

STEP 5: Alarm ON and LED glow

STEP 6: Else

STEP 7: Motor doesn't stop it goes on moving.

STEP 8: Alarm OFF and LED not glow

1.8 BLOCK DIAGRAM

The block diagram represents the alcohol sensing alert with motor locking project, illustrating the basic components used to create this device and their interconnections. Additionally, according to the code, the relay activates or deactivates the buzzer to alert the individual.

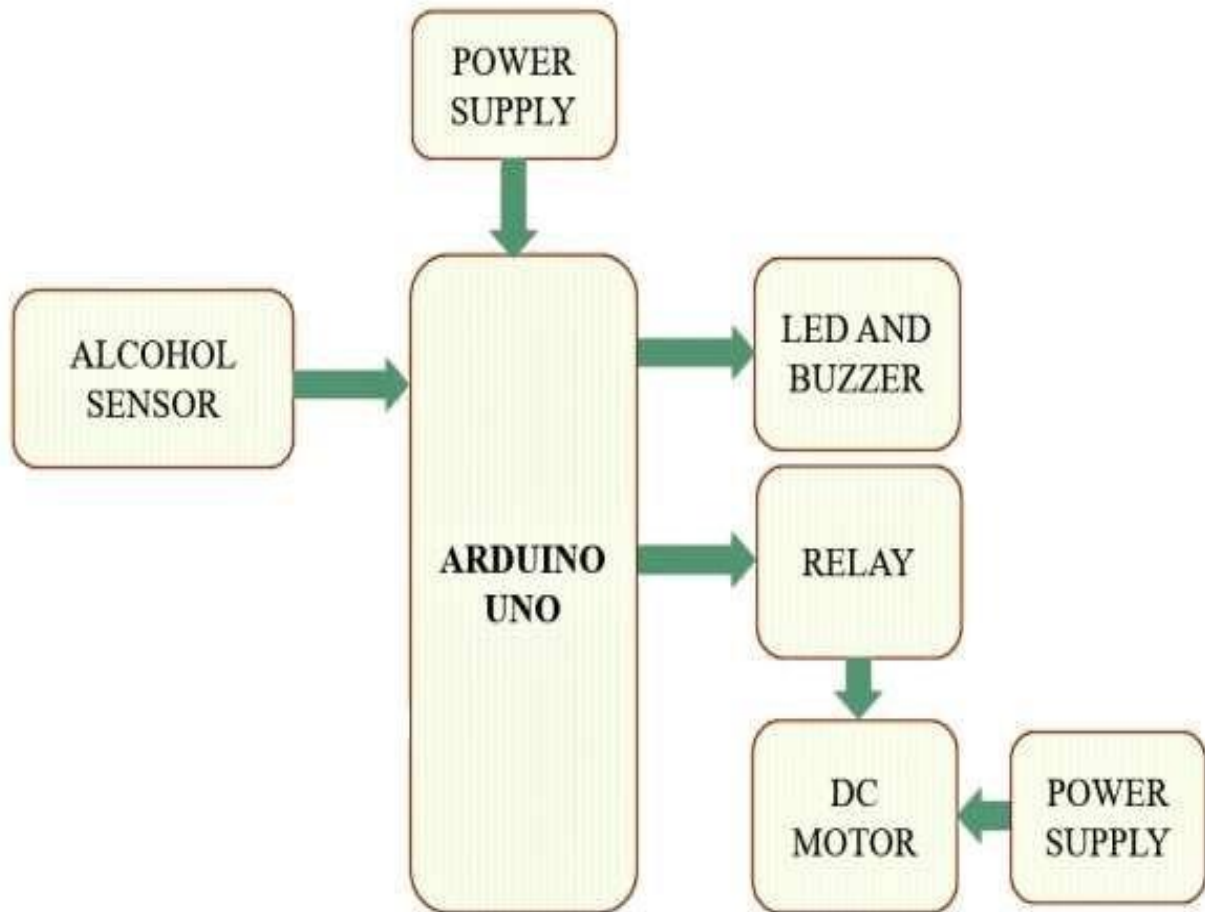


Fig.1.2: Block Diagram

Initially, the power supply and alcohol sensor are linked to the Arduino UNO. Upon detecting alcohol in the person's breath, the sensor sends a signal to the Arduino UNO. The relay, connected to the Arduino UNO, assists in halting the motor while activating the buzzer for alarm sound, and illuminating the LED for indication, thereby stopping the motor.

1.9 CIRCUIT DIAGRAM

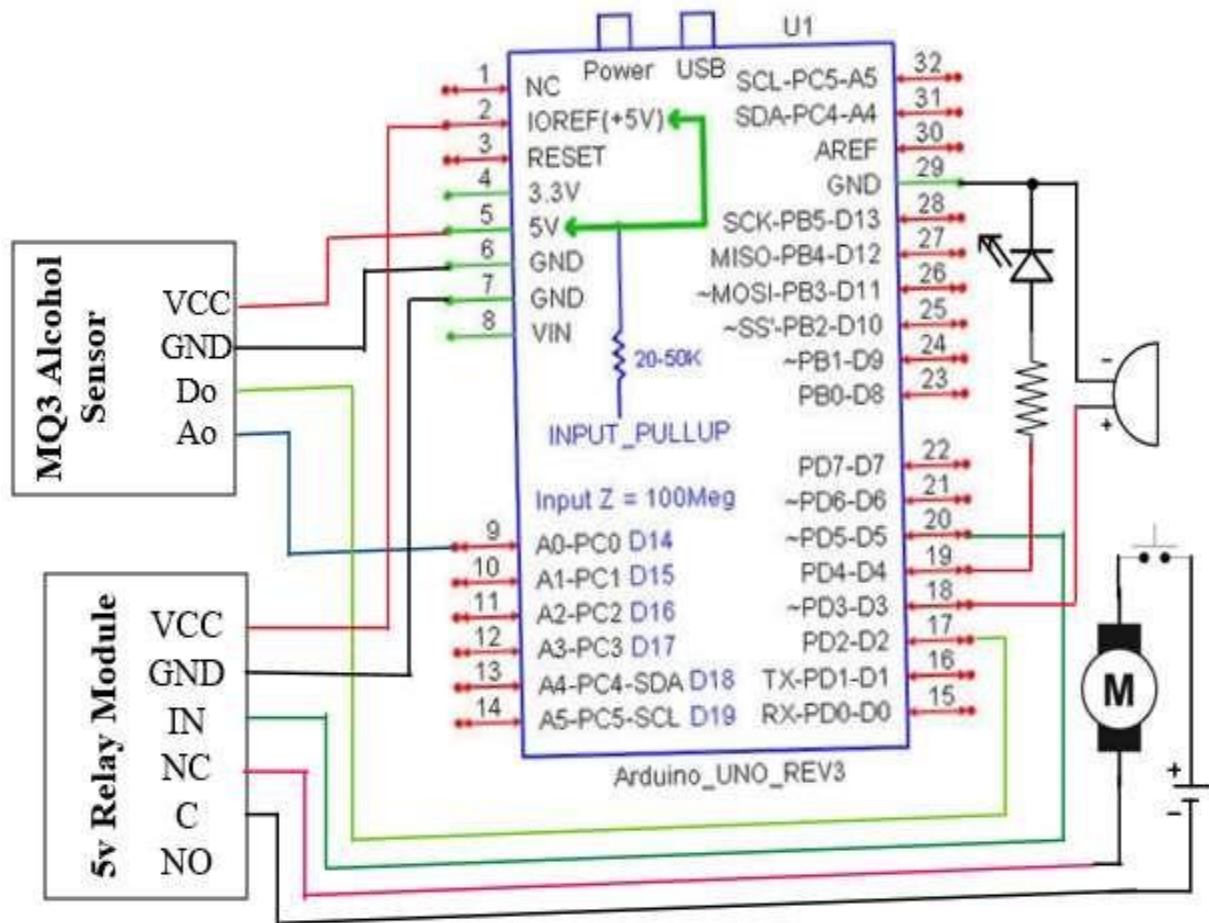


Fig.1.3: Circuit diagram

1.10 COMPONENTS USED

- ❖ Power Supply
- ❖ Arduino UNO
- ❖ MQ-3 Alcohol Sensor
- ❖ Buzzer and LED
- ❖ DC motor
- ❖ 5V Relay Module
- ❖ Resistor
- ❖ Jumper wire
- ❖ Push Button

Our system is powered with a 9V battery. A 5V DC supply as required by the microcontroller, and sensor. While other components like DC motor require 1.5V and the LED need 2V. The Arduino Uno board has already been designed to operate without the use of transformer, the system can be powered via the USB connection from computer or with an external power supply of 7 to 12V. The External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. Any voltage that is above 12V will make the control device to burn thereby destroying the board. It is advisable to use voltage between 7 - 12V

Arduino UNO is a low-cost, flexible, and easy-to-use programmable open-source microcontroller board that can be integrated into a variety of electronic projects. This board can be interfaced with other Arduino boards, Arduino shields, Raspberry Pi boards and can control relays, LEDs, servos, and motors as an output.

This board contains a USB interface i.e. USB cable is used to connect the board with the computer and Arduino IDE (Integrated Development Environment) software is used to program the board.

The unit comes with 32KB flash memory that is used to store the number of instructions while the SRAM is 2KB and EEPROM is 1KB.



The operating voltage of the unit is 5V which projects the microcontroller on the board and its associated circuitry operates at 5V while the input voltage ranges between 6V to 20V and the recommended input voltage ranges from 7V to 12V.

Arduino UNO Components: -

The Arduino UNO board contains the following components and specifications:

ATmega328: This is the brain of the board in which the program is stored.

Ground Pin: there are several ground pins incorporated on the board.

PWM: the board contains 6 PWM pins. PWM stands for Pulse Width Modulation, using this process we can control the speed of the servo motor, DC motor, and brightness of the LED.

Digital I/O Pins: there are 14 digital (0-13) I/O pins available on the board that can be connected with external electronic components.

Analogue Pins: there are 6 analogue pins integrated on the board. These pins can read the analogue sensor and can convert it into a digital signal.

AREF: It is an Analog Reference Pin used to set an external reference voltage.

Reset Button: This button will reset the code loaded into the board. This button is useful when the board hangs up, pressing this button will take the entire board into an initial state.

USB Interface: This interface is used to connect the board with the computer and to upload the Arduino sketches (Arduino Program is called a Sketch)

DC Power Jack: This is used to power up the board with a power supply.

Power LED: This is a power LED that lights up when the board is connected with the power source.

Micro SD Card: The UNO board supports a micro-SD card that allows the board to store more information.

3.3V: This pin is used to supply 3.3V power to your projects.

5V: This pin is used to supply 5V power to your projects.

VIN: It is the input voltage applied to the UNO board.

Voltage Regulator: The voltage regulator controls the voltage that goes into the board.

SPI: The SPI stands for Serial Peripheral Interface. Four Pins 10(SS), 11(MOSI), 12(MISO), 13(SCK) are used for this communication.

TX/RX: Pins TX and RX are used for serial communication. The TX is a transmit pin used to transmit the serial data while RX is a receive pin used to receive serial data.

III. MQ-3 Alcohol Sensor:

The sensor is made of Tin Dioxide (SnO_2) sensitive layer. The sensor is configured with a high sensitivity to alcohol and small sensitivity to Benzene. It has a simple drive circuit with fast response, stability, and long life. It has an analog interface type. On the sensor, port pins 1, 2 and 3 represents the output, GND and VCC respectively. The technical specification of the sensor is portrayed in table.

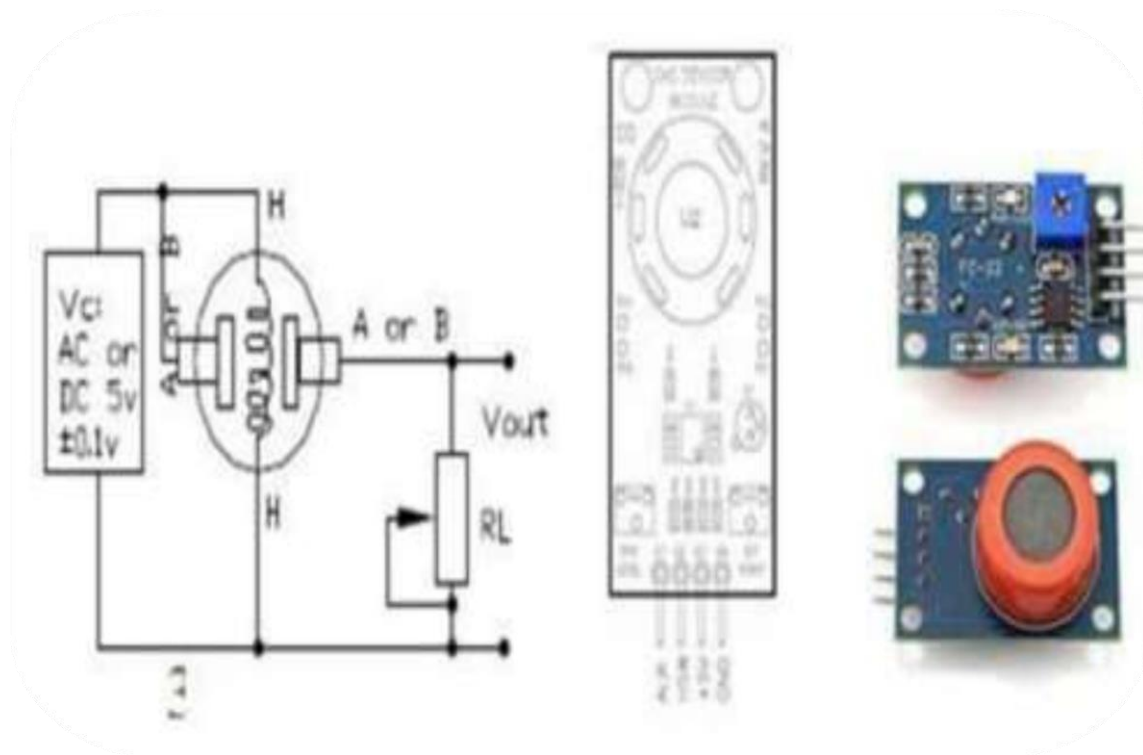


Fig.1.5:MQ-3 Alcohol Sensor

A. Standard work condition: -

Symbol	Parameter name	Technical condition	Remark
Vc	Circuit voltage	$5V \pm 0.1$	AC OR DC
VH	Heating voltage	$5V \pm 0.1$	AC OR DC
RL	Load resistance	$200K\Omega$	
RH	Heater resistance	$33\Omega \pm 5\%$	Room Tem
PH	Heating consumption	less than 750mw	

Table 1.1:MQ-3 Sensor Standard work condition

B. Environment condition: -

Symbol	Parameter name	Technical condition	Remarks
Tao	Using Tem	10°C-50°C	
Tas	Storage Tem	20°C-70°C	
RH	Related humidity	less than 95%Rh	
O ₂	Oxygen concentration	21% (standard condition) Oxygen concentration can affect sensitivity	minimum value is over 2%

Table 1.2:MQ-3 Sensor Environment condition

C. Sensitivity characteristic: -

Symbol	Parameter name	Technical condition	Remarks
Rs	Sensing Resistance	1M Ω - 8 M Ω (0.4mg/L alcohol)	Detecting Concentrations scope: 0.05mg/L - 10mg/L Alcohol
α (0.4/1 mg/L)	Concentration slope rate	0.6	
Standard detecting condition	Temp: 20°C $\pm$ 2°C Vc:5V $\pm$ 0.1 Humidity: 65% $\pm$ 5% Vh: 5V $\pm$ 0.		
Preheat time	Over 24 hours		

Table 1.3:MQ-3 Sensor Sensitivity characteristic condition

The circuit diagram of the MQ-3 sensor is shown in figure. In the datasheet, the recommended value to be used ranges from 100k ohm to 470k ohm. Here, 200k ohm was used.

IV. Alarm and Indicating Unit:

The alarm unit used is a buzzer which indicates when alcohol is detected. The buzzer used belongs to the PS series. The PS series are high performance buzzers that employ Uni-morph piezoelectric elements and are designed for easy incorporation into various circuits. They have very low power consumption in comparison to electromagnetic units. It has a voltage requirement of 2V and is connected to pin 10 of the microcontroller. The standard resistor value

of $220\ \Omega$ commercially available is closest to the computed value of $250\ \Omega$, so a $220\ \Omega$ resistor was used to limit the current going through the LED.

Buzzer:-

Fig.1.16: Buzzer

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric (piezo for short). Typical uses of buzzers and beepers include alarm devices, timers, train and confirmation of user input such as a mouse click or keystroke.

LED: -

A light-emitting diode is a semiconductor device that emits light when current flows through it. Electrons in the semiconductor recombine with electron holes, releasing energy in the form of photons. The color of the light is determined by the energy required for electrons to cross the band gap of the semiconductor.

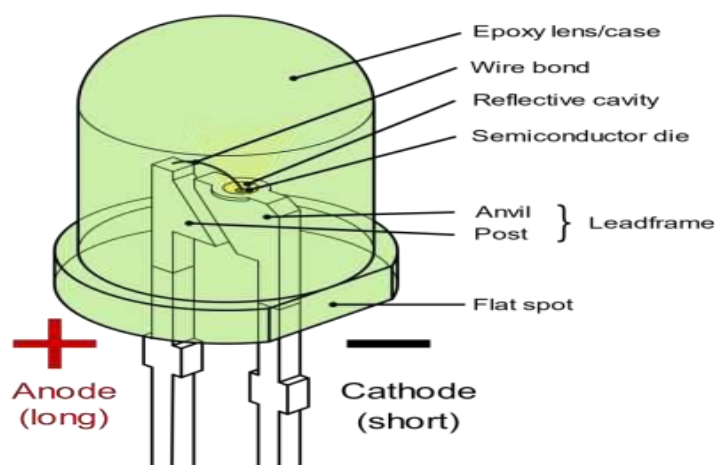


Fig.1.17: LED

LEDs are so common, they come in a huge variety of shapes, sizes and colours. The LEDs you are most likely to use are the standard through hole LEDs with two legs. Following figure shows the parts of it.

Like an ordinary diode, the LED operates only in forward bias condition. When the LED is forward biased, the free electrons cross the PN junction and recombine with holes. Since these electrons fall from a higher to a lower energy level, they radiate energy in the form of photons (light).

In ordinary diodes, this energy is radiated as heat while in an LED, energy is radiated as light. This effect is called Electroluminescence.

V. DC motor

This 5V DC motor is a type of direct current (DC) motor that operates at a voltage of 5 volts. 5v high speed dc motor is a small, low-power motor that is often use in toys, hobby projects, and other applications where a small and lightweight motor needed.

The 5V mini-DC motors are typically made of plastic, metal and have a small shaft that can use to connect the motor to other devices.

The speed of a PMMC mini toy motor is determine by the voltage applied to it. The higher the voltage, the faster the motor will rotate. The torque of a motor is its ability to turn against a load. The higher the torque, the more force the motor can apply to the load.

PMMC mini toy 5V DC motors are typically rated for a maximum current of 100 milliamperes (mA). This means that they can draw a maximum of 0.1 amps of current. The actual current draw of a motor will vary depending on the load that is applied to it.

The working principle of a 5V DC mini toy motor is based on the interaction between a magnetic field and an electric current. The motor has two main parts: a stator and a rotor. The stator is the stationary part of the motor. It has a set of permanent magnets that create a magnetic field. The rotor is the rotating part of the motor. It has a coil of wire that is wound around a core.

5V PMMC Mini Toy Motor is a simple and small DC motor with low voltage and high RPM. This DC Motor is useful for different robotics and momentary projects. It can run in 3-6V supply and RPM can reach up to 14000.



Fig.1.8:5V DC motor

Features: –

Overall, 5v mini toy motors offer a number of benefits that make them a popular choice for a variety of applications. If you are looking for an inexpensive, efficient, and reliable motor, a PMMC mini toy motor is a good option to consider.

5V DC Motor Specifications: –

5v high speed dc motors are relatively inexpensive, making them a cost-effective option for many applications. small 5v dc motors are relatively simple to construct, making them a good choice for DIY projects. PMMC mini toy motors are efficient and can operate at low voltages, making them a good choice for battery-powered applications.

Standard 130 Type DC motor

Operating Voltage: 4.5V to 9V

Recommended/Rated Voltage: 6V

Current at No load: 70mA (max)

No-load Speed: 9000 rpm

Loaded current: 250mA (approximately)

Rated Load: 10g*cm

Dimensions: 20 x 15 x 25

Axis length: 8mm

Shaft diameter: 2mm

Weight: 015 kg

5V DC Motor Applications: -

Air compressor.

Winching system.

Electric traction.

Hair drier.

Vacuum cleaner and in speed regulation application.

VI. 5V Relay Module

A 5V relay module is a single or multi-channel relay module that works with a low-level trigger voltage of 5V DC. The input voltage can be from any microcontroller or logic chip that outputs a digital signal.

Like most other relays, the 5V relay module is an electrically operated, electromagnetic switch that can be used to turn on or turn off a circuit. It consists of two parts: the relay itself and the control module.

The relay contains the coil that creates the magnetic field, the armature that move to complete or disconnect a circuit, and contacts that open and close to operate the load switch.

The relay control module is the interface or part of the relay module that the user interacts with. It contains the input terminals for connecting to the microcontroller, as well as the output terminals for connecting to the load.

The control module also contains LED indicators for power and status and other devices such as protection diode, transistor, resistor, and other semiconductor devices necessary for its operation.

5V Relay Module Specifications:

The 5V relay module specifications are normally written on the top side. These include the input voltage/current, load current/voltage, and operating or release times. The specifications vary depending on the manufacturer. In general, they would look something like this:

Normal voltage: 5V DC

Normal current: 70mA

Maximum load current: 10A/250V AC, 10A/30V DC

Maximum switch voltage: 250V AC, 30V DC

Operate time: $\leq 10\text{ms}$

Release time: $\leq 5\text{ms}$

5V Relay Module Circuit:

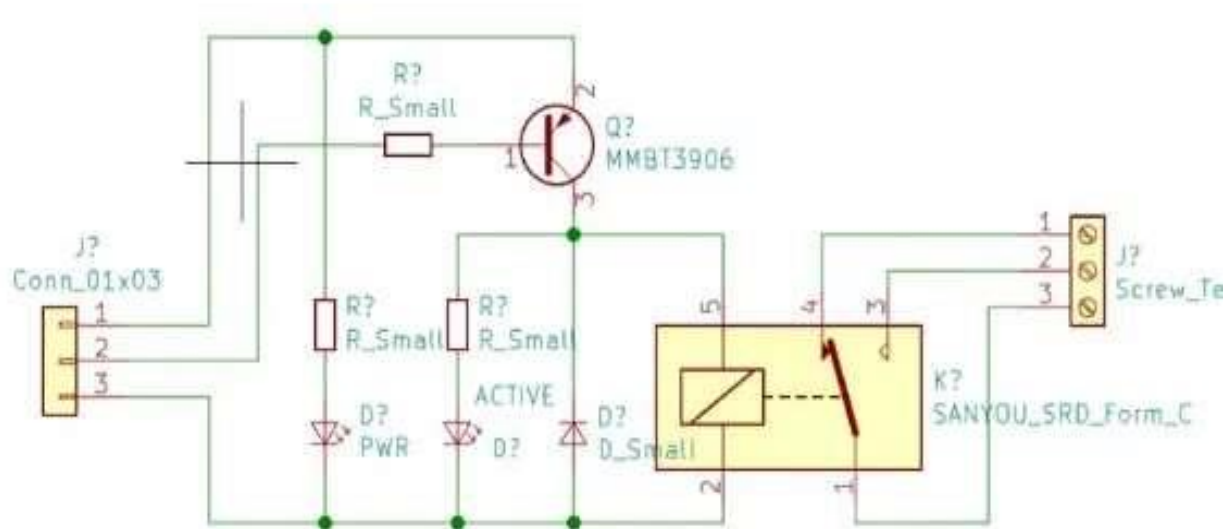


Fig.1.9: Relay module schematics

The 5V relay modules are made up of connection points or pins, and several major components, such as diodes, transistors, resistors, and the relay itself. These make up the circuit that controls the relay. The 5V relay module circuit is further explained below.

5V Relay Module Pinout:

The 5V relay module pinout is composed of connections on the input side where it receives the trigger signal, and the output side where it controls the load.

Input side:

The relay module circuit diagram as shown in the above, has 3 or 4 connections: These are listed and explained below.

- VCC – this is the power connection. It supplies 5V DC to the module and is normally connected to the positive terminal of the power supply.

- GND – this is the ground connection. It connects to the negative terminal of the power supply.
- IN1, IN2 – these are the inputs where the trigger signal is applied. IN1 is for a singlechannel relay module, while IN2 is for a dual-channel relay module. The IN (Input) pin is connected to the output of the microcontroller, sensor, or logic device.

The relay module output side has three connections:

- NO (Normally Open) – this is the load connection when the relay is ON. When the relay is off, the NO maintains an open connection with the COM.
- COM (Common) – The relay module connection labeled “COM” is the common connection for both the NO and NC (Normally Closed) pins.
- NC (Normally Closed) – this is the load connection. It connects to the COM terminal by default, or when the relay is OFF.

5V Relay Module Parts

Now, let's take a look at the main components of a 5V relay module. They include:

- The LED – this is an indicator or status LED that lights up when the relay is ON.
- The transistor – the transistor amplifies the trigger signal so that it can activate the relay.
- The diode – a flyback diode is used to protect the 5V relay module circuit from flyback voltage spikes when the relay coil de-energizes.
- The resistor – the resistor limits the current flowing through the relay module circuit.
- The relay – this is the main switching component of the module and usually either an NC (Normally Closed) or NO (Normally Open) type.



Fig.1.10:5V Relay 5V

Relay Module for Arduino:

The 5V relay module can be easily interfaced with an Arduino. The above diagram shows how to connect a single-channel relay module with Arduino microcontrollers, a popular project among hobbyists. The 5V relay module Arduino wiring is very simple:

You only need to connect the VCC and GND pins of the relay module to the 5V and GND pins of the Arduino, connect the IN1 pin of the relay module to a digital output pin of the Arduino, and connect the load to the NO and COM pins of the relay module.

When the digital output pin is set to HIGH, the module turns ON and activates the relay. This will turn ON the load connected to the NO and COM pins of the relay module.

When the digital output pin is set to LOW, the circuit OFF and deactivates the relay, turning OFF the load.

The 5V relay module can also be easily interfaced with a Raspberry Pi. When used with a Raspberry Pi, 5V relay module can be used to control a load such as an LED system, motor, or solenoid.

Applications of 5V Relay module:

If it is use in electronics projects, the 5V relay module is a very popular module among hobbyists. It is also widely used by professionals in home, industrial and commercial power systems and equipment. Common applications include:

- Home automation
- Robotics
- Light control
- Motor control
- Solenoid control
- Valve control
- Process control
- Alarm systems
- Access control systems
- Surveillance systems
- DIY projects

VII. Resistor

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to divide voltages, bias active elements, and terminate transmission lines, among other uses. High-power resistors that can dissipate many watts of electrical power as heat may be used as part of motor controls, in power distribution systems, or as test loads for generators. Fixed resistors have resistances that only change slightly with temperature, time or operating voltage. Variable resistors can be used to adjust circuit elements (such as a volume control or a lamp dimmer), or as sensing devices for heat, light, humidity, force, or chemical activity.



Fig.1.11: Resistor

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in electronic equipment. Practical resistors as discrete components can be composed of various compounds and forms. Resistors are also implemented within integrated circuits.

The electrical function of a resistor is specified by its resistance: common commercial resistors are manufactured over a range of more than nine orders of magnitude. The nominal value of the resistance falls within the manufacturing tolerance, indicated on the component.

A 330 Ohm resistor can be identified via resistor colour codes of Orange-Orange-BrownGold or Orange-Orange-Black-Black-Gold.

VIII. Jumper wire

A jumper wire is an electric wire that connects remote electric circuits used for printed circuit boards. By attaching a jumper wire on the circuit, it can be short-circuited and short-cut (jump) to the electric circuit.

Types of Jumper Wires: -

Jumper wires typically come in three versions: male-to-male, male-to-female and female-to-female

Male to Male jumper wires: -

These are male to male jumper wires used in connecting the male header pin of any development board to other development boards having a male connector.

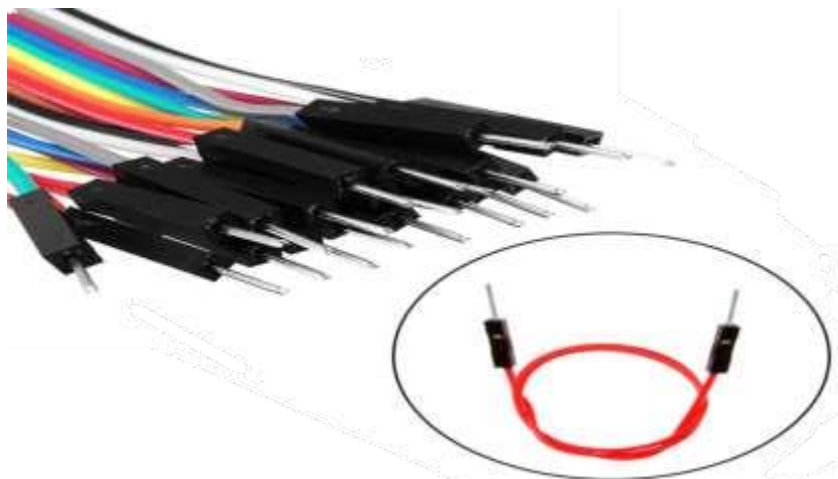


Fig.1.12: male to male jumper wires

They are simple wires that have connector pins at each end allowing them to be used to connect two points to each other. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed. Male to Male jumper wire each cable length about 20cm or 8-inch.

Male to Female jumper wires: -

Male to female jumper wires is an essential component for any electronic prototyping and testing project. They allow for easy and convenient interconnection between components without the need for soldering. These wires come in groups or cables with connectors or pins at each end and are commonly used for connecting FRC pins, Header pins, Berg pins, and other components.

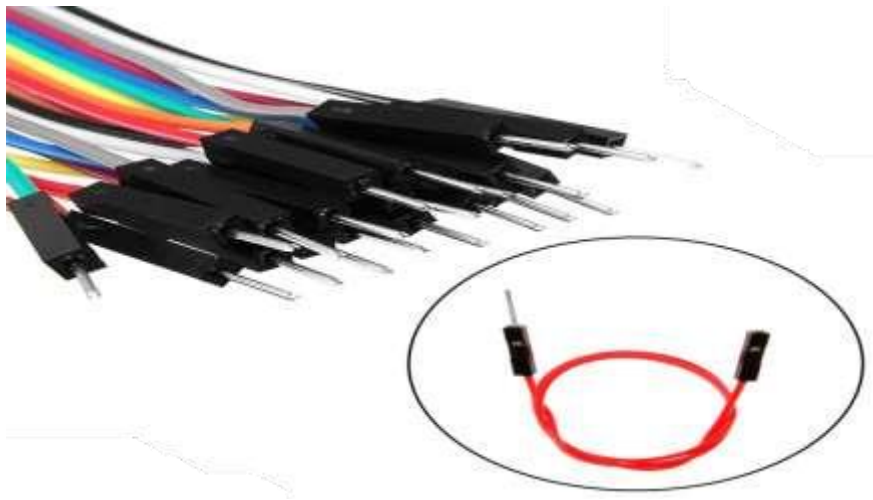


Fig.1.13: male to female jumper wires

The male to female jumper wires, in particular, have a male pin on one end and a female connector on the other, making them suitable for a wide range of projects that require connecting components with different types of pins or connectors. One popular use for these jumper wires is with microcontrollers such as the Arduino. These boards have a series of pins that control various components and sensors, and with the help of male to female jumper wires, these pins can easily be connected to other components on a breadboard or in a circuit.

In addition to Arduino projects, male to female jumper wires is also useful for connecting components on a PCB project or a PC motherboard. Their easy plug-and-play design makes it possible to test different component configurations quickly without the need for permanent connections. This allows for efficient troubleshooting and design refinement before making a final, permanent connection.

Features: -

It connects two points to each other without soldering

It is reusable

It is inexpensive and easy to use

Applications: -

It is used to interconnect the components of a breadboard or other prototype or test circuit internally or with other equipment or components without soldering.

IX. Tactile Switch or Push Button

Tactile Switches, also known as push buttons, can be found everywhere: from phones to elevators to large industrial machines. They are the simplest type of input device.



Fig.1.14: Push Button

A push button helps in closing an open electronic circuit or the reverse temporarily during the time it is pressed. It has a small spring which helps it to return to its original position when the user takes the pressure off the button.

There are mainly two types of push-buttons:

1. Normally open (NO)
2. Normally closed (NC).

Normally open (NO) Push Button: -

A circuit with a normally open push-button is open by default. When the user presses the button, the circuit becomes complete and the current starts flowing through it.

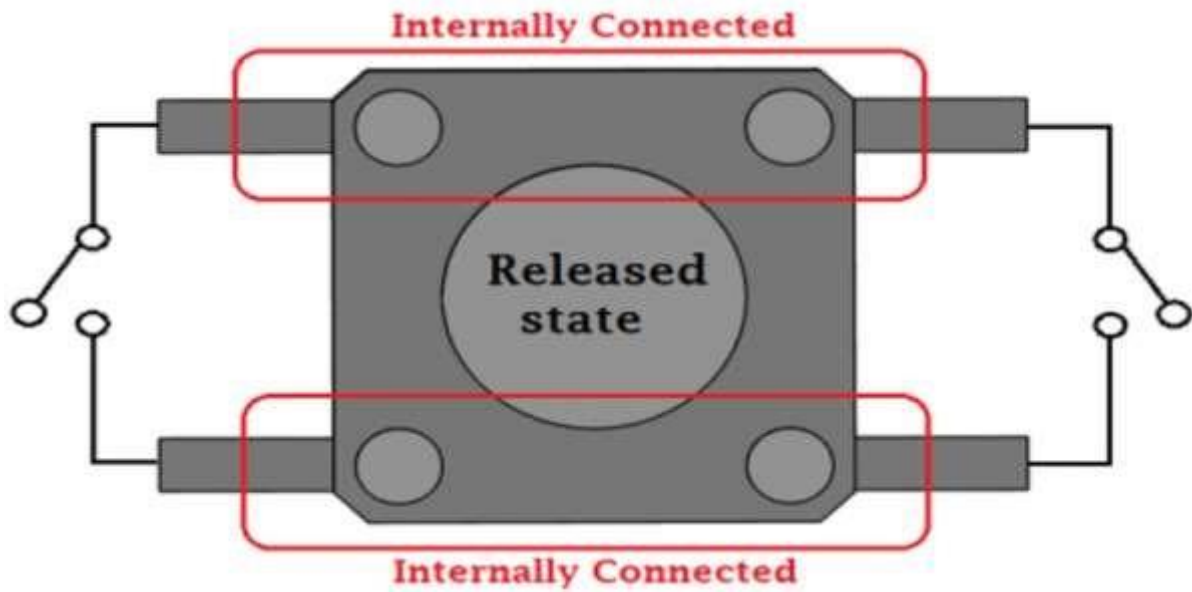


Fig.1.115: Push Button Pinout/Connections in Released state

Normally closed (NC) Push Button: -

A circuit with a normally closed push-button is closed by default. The circuit breaks on pressing the button.

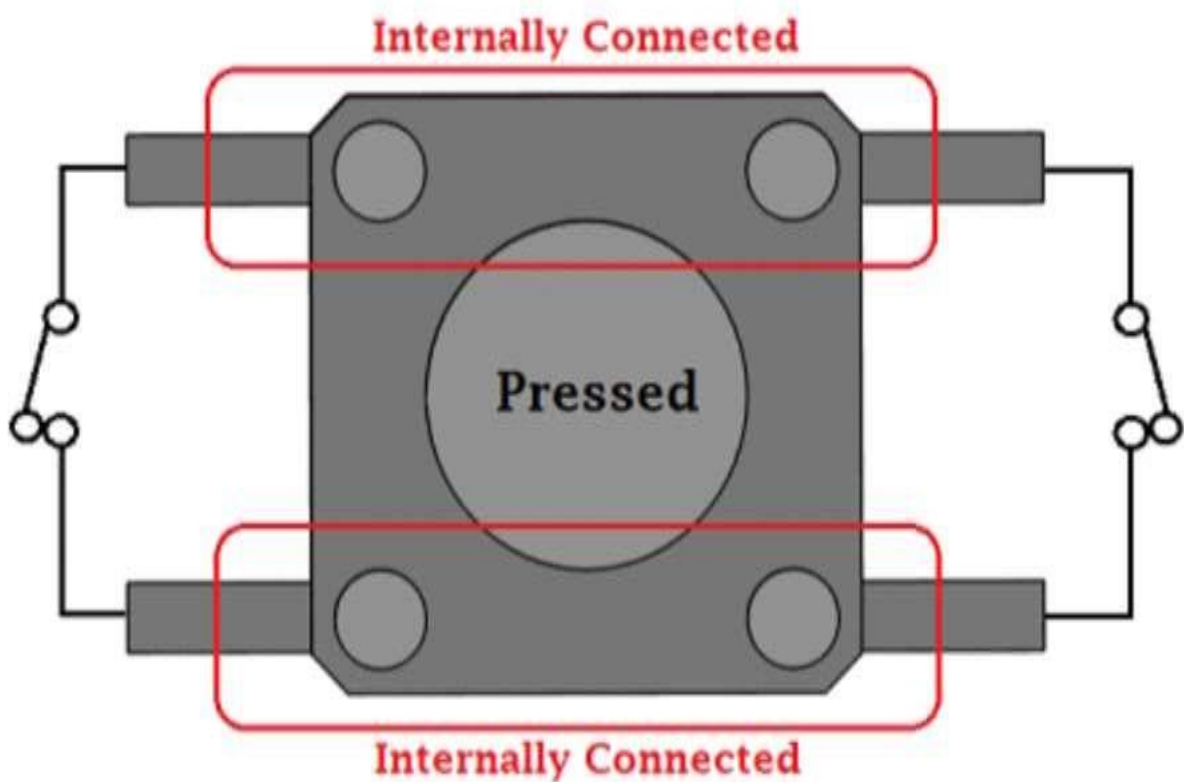


Fig.1.16: Push Button Pinout/Connections in Pressed state

Specifications: -

Operating Voltage	12 V DC
Operating Current	50mA
Switch Type	Momentary
Contact Resistance	30mΩ
Insulation Resistance	100MΩ
Operating Temperature Range	-30 to 70 deg C
Length (mm)	6
Width (mm)	6
Height (mm)	5

Table 1.4: Push Button Specifications

Features: - ○ Through-hole design. ○ Shaft Shape: Round ○ The long life of

300,000 cycles realized despite the high operating force ○ Used in the fields of electronic products, household appliances and more.

- This light touch switch is waterproof, prevent oil, anti-pollution, anti-static interference.
- High precision mechanism design offers acute operation and long service life. ○ Compact and lightweight, easy to carry and dismantling.
- Good electrical conductivity.

Applications: -

- Can be used in Prototyping
- Can be used in DIY Projects for triggering input.
- Can be used with Arduino for IOT Applications

1.11 PROCEDURE FOR UPLOAD A CODE

1. Install Arduino software in your system.
2. Make your code and save it as .into file and compile it successfully.
3. After making all connections in the circuit, we will connect the Arduino using USB cable to our system.
4. We will then upload the code and will print the required results in serial window.

1.12 SOURCE CODE

```
#define sensor Digital 2

#define LED 3

#define buzzer 4

#define sensorAnalog A1

#define relay 5

void setup ()

{

pinMode(sensorDigital, INPUT);

pinMode(LED, OUTPUT);

pinMode(buzzer, OUTPUT);

pinMode(relay, OUTPUT);

Serial.begin(9600);

}

void loop ()

{

bool digital = digitalRead(sensorDigital);

int analog = analogRead(sensorAnalog);

Serial.print("Analog value : ");

Serial.print(analog);

Serial.print("\n");

Serial.print("Digital value :");

Serial.println(digital);

if (digital == 0)

{

digitalWrite(LED, HIGH);

digitalWrite(buzzer, HIGH);

digitalWrite(relay, LOW);

}

else

{
```

```
digitalWrite(LED, LOW);  
digitalWrite(buzzer, LOW);  
digitalWrite(relay, HIGH);  
}  
}
```

1.13 RESULT

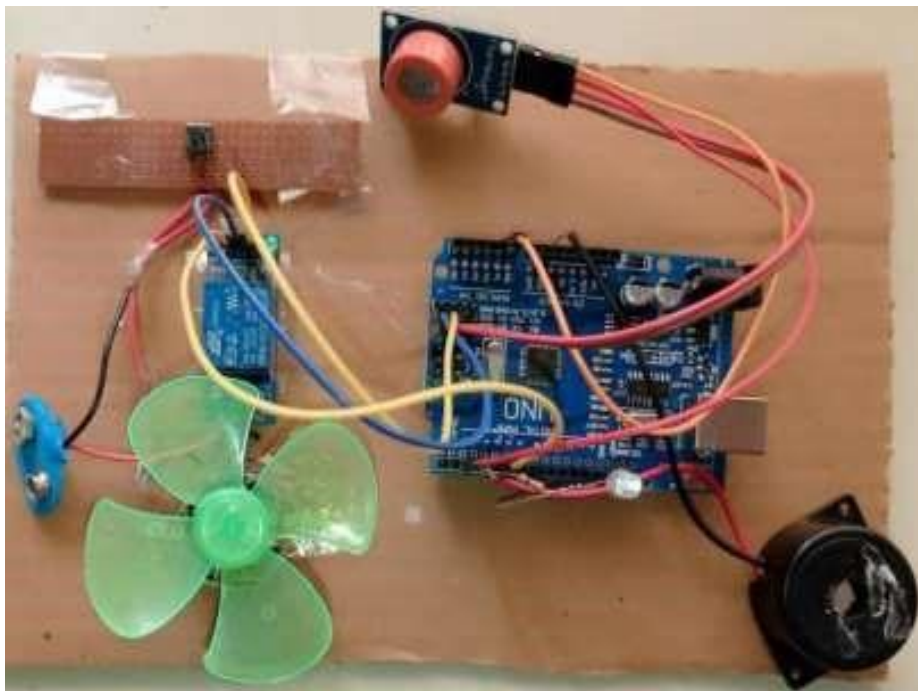


Fig.1.17: System is off state

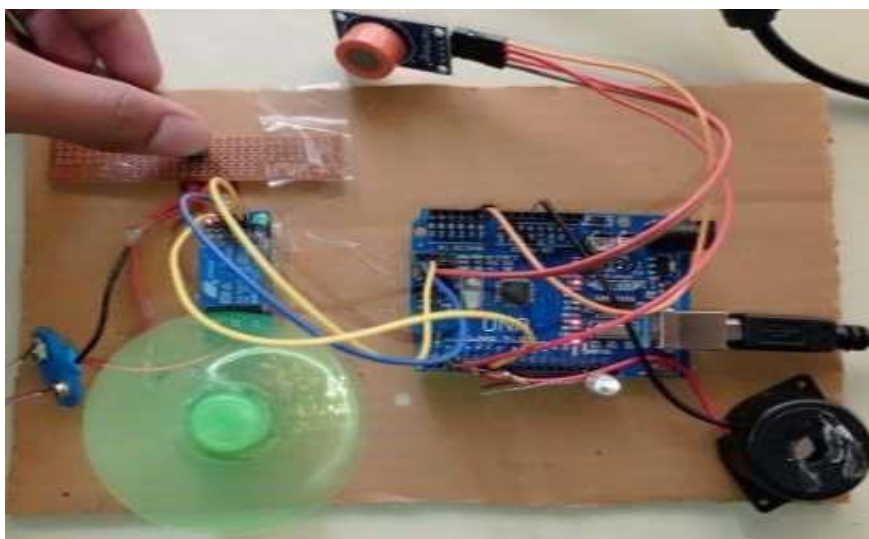


Fig.1.18: System ON state without Alcoholic data

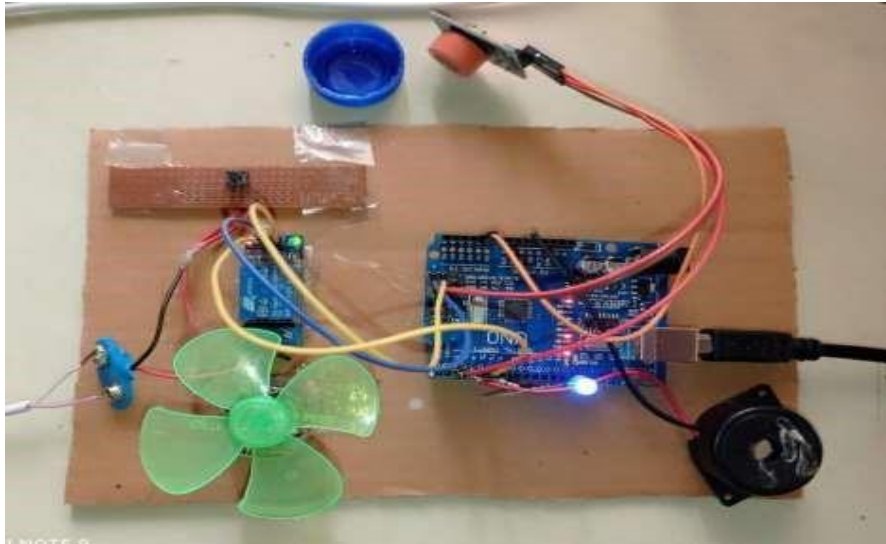


Fig1.19: System ON state with Alcoholic data

- When a person under the influence of alcohol attempts to operate the vehicle, the alcohol sensor immediately detects the presence of alcohol. Upon detection, the sensor triggers the shutdown of the motor and emits an alarm, alerting nearby individuals.
- This prompt alert enables bystanders to recognize the situation and take appropriate action as needed, mitigating potential risks.
- Thus, the integration of this system in vehicles helps prevent any loss of life or property damage.
- Comprehensive testing and proper connection of all components ensure the system functions effectively, as depicted in the image above.

1.14 APPLICATIONS

1. Ignition Interlock Systems: - Prevent a vehicle from starting if the driver's alcohol level is above the legal limit. This application is commonly used to reduce drunk driving incidents.
2. Fleet Management: - Implemented in commercial vehicles and fleets to ensure that drivers are not under the influence of alcohol while operating vehicles, enhancing road safety and compliance with transportation regulations.
3. Public Transportation Safety: - Deployed in public transportation systems, such as buses or taxis, to ensure that drivers are sober when transporting passengers.
4. Rental Car Security: - Integrated into rental car systems to enhance security and prevent individuals under the influence from renting and driving vehicles.

5. Emergency Response Vehicles: - Ensure that drivers of emergency vehicles (ambulances, fire trucks, etc.) are not impaired, guaranteeing the safety of both the emergency responders and the public.
6. Corporate Fleets: - Used by companies with vehicle fleets to promote responsible driving behavior among employees, ensuring the safety of company vehicles and reducing liability.
7. School Buses: - Implemented in school buses to ensure the safety of students by preventing drivers from operating the bus under the influence of alcohol.
8. Rideshare Services: - Integrated into rideshare platforms to enhance passenger safety by ensuring that drivers are not impaired.
9. Law Enforcement Vehicles: - Used in police vehicles to ensure that law enforcement officers are not under the influence while on duty, promoting responsible behavior and public safety.
10. Personal Vehicles: - Adopted as a safety feature in personal vehicles to prevent individuals from driving under the influence, reducing the risk of accidents and legal consequences.
11. Insurance Telematics: - Integrated into insurance telematics systems to monitor driver behavior, potentially leading to reduced insurance premiums for responsible drivers.
12. Custom Vehicle Modifications: - Incorporated into custom vehicle modifications for specific applications where access to the vehicle needs to be restricted based on the driver's alcohol levels.

1.15 ADVANTAGES

- The Alcohol detection with Motor locking system can be implemented in any 4-wheelers
- The Alcohol detection with Motor locking system can help prevent accidents due to drunk driving.
- The Alcohol detection with Motor locking system can be very helpful for police.
- The Alcohol detection with Motor locking system provide automatic safety system for cars and other vehicles
- The chance of loss of life and property due to drunken driving is minimized.
- Simple implementation leads to accurate results.

- Can be implemented on various types of vehicles.
- Less accidents, more safety.

1.16 DISADVANTAGES

- It requires some extra amount to implement
- Implementation is difficult, but it can be implemented in inbuilt
- Damage of sensor cannot be detected

1.17 FUTURE SCOPE

We can implement GSM technology to inform the relatives or owner of the vehicle about the alcohol consumption.

1.18 VIDEO LINK: - <https://youtu.be/bEoSbPHEYmE?si=6Mo3clRiHSrNvAml>

CHAPTER-2

AUTO ELECTRIC VEHICLES

2.1 INTRODUCTION



Fig.2.1: Entrance

The Auto EV India 2023, trade show, exclusively dedicated to the entire eMobility ecosystem, more particularly with EV Technologies, brings the entire Electric Vehicle value chain under one roof to network with major industry players and identify numerous business opportunities in a hypercompetitive EV market.

Auto EV India 2023, an EV Technology Exhibition, broadly focuses on Drive Systems, Rechargeable next-generation Batteries, Motor Technologies, Parts and Materials, Microcontrollers, Driver ICs, Control Modules, Software, Sensors, Measurement and Simulation, Production Facilities, Inverters, Peripherals, Converters, Power Devices, Passive Elements, Heat-Resistant Products, Inverter Evaluation and Testing System, Chargers, Connectors, and Harnesses. The event also aims to enable a large showcase of the new EV models as innovative vehicle solutions.

The Indian Automotive industry is increasingly absorbing new technologies and promoting indigenous R&D for adapting these technologies for Electric Vehicles. Auto EV India paves the way for the newest components and technologies in the realm of eMobility and opens the window for technology transfers. Auto EV India offers the ideal platform to showcase new products and technologies and new product launches, concept vehicles, trade dialogues, and huge overseas participation.

2.2 The Tata Nexon Electric Vehicle



Fig.2.2: Tata Nexon

Electric cars are vehicles powered by electric motors, which are fueled by rechargeable battery packs. They are becoming increasingly popular due to their environmental benefits, lower operating costs over time, and advancements in battery technology, which have extended their range and reduced charging times. Major automakers are investing heavily in electric vehicle development, aiming to make them more accessible and competitive with traditional internal combustion engine vehicles.

Tata Nexon is an electric vehicle (EV) model produced by Tata Motors, an Indian automotive manufacturer. The Tata Nexon EV is an all-electric SUV that is part of Tata's commitment to sustainable mobility and reducing the environmental impact of transportation. It was one of the early electric SUVs to be launched in India.

1. **Ziptron Technology:** Tata's Ziptron technology is the heart of the Nexon Electric. It features a powerful electric motor and a high-capacity lithium-ion battery that provides a long driving range.
2. **Long Range:** The Nexon Electric offers a range of over 300 kilometers on a single charge, which is a significant achievement in the affordable electric SUV segment.
3. **Fast Charging:** It supports fast charging, allowing you to charge up to 80% of the battery in just 60 minutes, making it convenient for daily use.
4. **Connected Features:** The Nexon Electric comes equipped with advanced connected car features, enabling remote vehicle monitoring, diagnostics, and over-the-air updates.

5. Regenerative Braking: It features regenerative braking, which recovers energy during braking and deceleration to increase the vehicle's efficiency.
6. Safety Features: Equipped with standard safety features like ABS with EBD, dual airbags, and stability control.
7. Infotainment System: Comes with a touchscreen infotainment system that supports smartphone integration, navigation, and more.

2.3 7KW ON BOARD BATTERY CHARGER MODULAR SYSTEM 1-PHASE OR 3-PHASE SOLUTIONS ALLOWING UP TO 21KW



Fig.2.3: Battery charger modular

Modular battery chargers represent a versatile and adaptable solution for charging multiple batteries simultaneously. Unlike traditional chargers with fixed charging configurations, modular chargers offer flexibility and scalability to accommodate diverse battery types, sizes, and charging requirements.

1. Dynamic Load Balancing: Implement an intelligent load balancing system that can distribute power dynamically among multiple chargers in the same location. This ensures efficient utilization of available power and avoids overloading the electrical grid.
2. V2G (Vehicle-to-Grid) Integration: Enable bidirectional power flow, allowing electric vehicles to not only charge from the grid but also provide surplus energy back to the grid during peak demand periods, creating a more resilient and sustainable energy ecosystem.
3. Wireless Charging: Explore wireless charging technologies, such as inductive or resonant charging, to eliminate the need for physical connectors, making the charging process more convenient and reducing wear and tear on charging cables.

4. **Advanced User Interface:** Develop a user-friendly interface with features like smartphone app integration, real-time charging status updates, and predictive maintenance alerts for a seamless and interactive user experience.
5. **Energy Storage Integration:** Include an option for integrating energy storage solutions like home batteries or stationary storage systems to store excess energy for later use, enhancing energy independence and reliability.

2.4 GODAWARI ELECTRIC MOTORS PVT LTD



Fig2.4: Electric scooters

Electric scooters, often abbreviated as e-scooters, are two-wheeled vehicles propelled by electric motors. They have gained significant popularity in recent years due to their convenience, eco-friendliness, and suitability for urban commuting.

1. Godawari Electric Motors Private Limited is a young and innovative company with vision and mission to provide the best quality products and best in class service and spare facilities.

Vehicle coming from godawari motors are all non-polluting vehicle, which intern help a better society.

2. All the products Manufactured by Godawari Electric Motors are designed from scratch keeping in mind, the consumer aspiration and needs focusing on safety comfort and performance. The innovative streak can be seen in all our products.

3. To cater the world class products and services Godawari Electric Motors has built a Hi-tech infrastructure with a manufacturing unit covering 1,00,000 square feet setup in Raipur (C.G).

4. Our product life spans from e-cycle to e-auto covering the complete the largest EV-range in India. All these vehicles are fitted with lithium-ion batteries which guarantees best mileage, safety and warranty.

2.5 LIRON LIB POWER PTE LTD



Fig.2.5: Lithium ion batteries

A lithium-ion or Li-ion battery is a type of rechargeable battery that uses the reversible intercalation of Li^+ ions into electronically conducting solids to store energy. In comparison with other commercial rechargeable batteries, Li-ion batteries are characterized by higher specific energy, higher energy density, higher energy efficiency, a longer cycle life, and a longer calendar life.

1. LiRON LIB Power is a Singapore based Hi-Tech Company focusing on rechargeable Lithium-Ion batteries research, design, development and manufacturing. Batteries developed and produced by LiRON focuses on high cycle life, fast charging, higher discharge current and good safety performance that also works well at low temperature and with high temperature tolerance.

2. These features make the battery cells ideal for applications in power tools, portable home appliances, E-bike/E-scooter and EV.

3. Another area is in the Energy Storage System (ESS) for solar, wind and other renewable energy applications where wide temperature range and safety are most desirable.

4. LiRON also provides technical assistance and support to partners establishing battery packs assembly operation.

2.6 I-3 Sensopart



Fig.2.6: vision sensor

A vision sensor is a device that captures and processes visual information from the environment, mimicking human vision to detect, analyze, and interpret images or scenes. These sensors utilize various technologies such as cameras, lenses, and image processing algorithms to gather data and extract meaningful insights. They find applications in diverse fields like robotics, automation, surveillance, automotive safety, and quality control, enabling machines to perceive and interact with their surroundings more intelligently and efficiently

A high-performance smart camera fitted in a compact and light-weight housing forms the heart of our VISOR[®] vision sensors. They feature innovative solutions, such as a target laser for simple mechanical alignment or motorised focus for automatic adjustment to your individual operating distance.

The hardware is optimally enhanced by the VISOR[®] software, which enables the set up of applications in a few simple steps and the adjustment of process parameters. Thanks to the software, the process can also be continuously monitored.

Sensopart vision sensors have proven themselves to be extremely flexible – the VISOR[®] vision sensor comes with diverse resolutions of up to 5 megapixels and three different fields of view (narrow to wide). One variant for special applications allows the additional installation of an external lens.

2.7 HELLERMANN TYTON

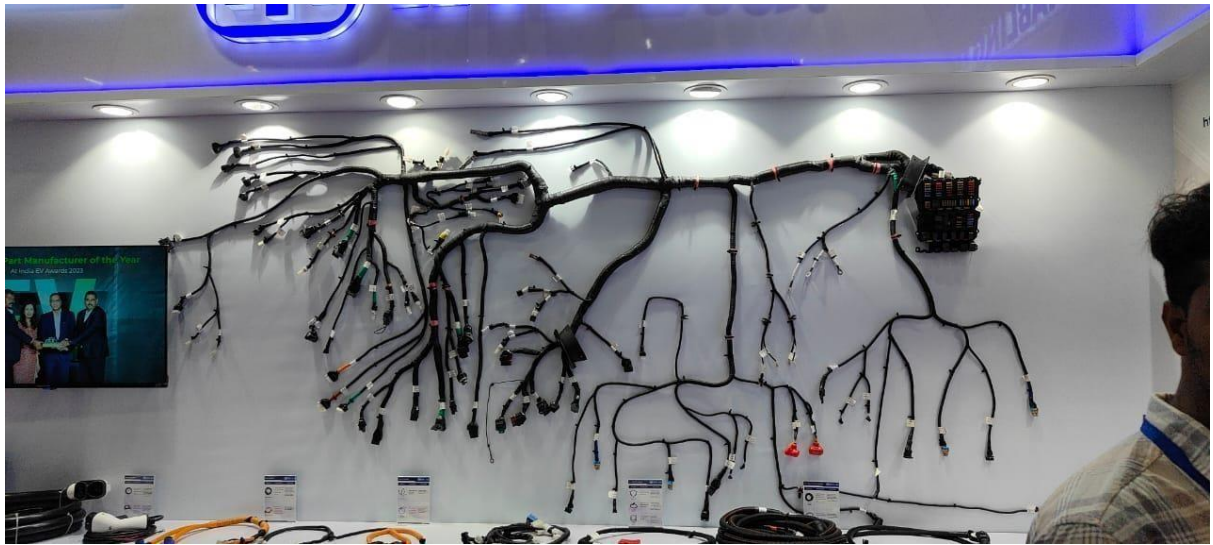


Fig2.7: Cables

Cables are physical conduits used to transmit electrical signals, data, or power from one point to another. They come in various types, including coaxial, twisted pair, fiber optic, and USB, each designed for specific purposes and environments. Cables play a crucial role in connecting devices, networks, and systems, facilitating communication and power distribution across different devices and locations.

ABOUT THE COMPANY: HellermannTyton is a leading manufacturer and supplier of quality products for clamping, installing, connecting, insulating, protecting and identifying electrical cables with efficient connection solutions. We also develop cable ties according to the specific requirements of the industry

Company mission is to develop efficient solutions for all cable ties management requirements and that in order to achieve this goal, we have to understand everything that happens in the world of our customers and, as this is the main inspiration for the development of our product groups, we have 18 world-class manufacturing factories and product development centers in 13 countries, with more than 6,000 employees working for Hellermann Tyton in 39 countries.

We also have the first-class services and advice for providing to you wherever you need them.

1. Sustainable Materials: Explore the use of sustainable materials and manufacturing processes to reduce the environmental footprint of their products. This includes developing cable management solutions using recycled or eco-friendly materials.

2. 2.IoT Integration: Incorporate Internet of Things (IoT) technology into their cable management products, allowing real-time monitoring and data collection for better cable management and maintenance.
3. 3.3D Printing Customization: Offer 3D printing options for customized cable management solutions, allowing customers to design and print cable clamps and accessories tailored to their specific needs.
4. 4.Augmented Reality (AR) Support: Develop an AR app or platform that assists technicians in the installation and maintenance of cable management systems, providing step-by-step visual guidance.
5. 5.Smart Cable Identification: Create innovative cable identification and labeling systems that can be scanned with a smartphone to access information about the cable, making troubleshooting and maintenance more efficient.
6. 6.Digital Twins: Implement digital twin technology to create virtual replicas of cable management systems, enabling remote monitoring, predictive maintenance, and easier modifications.



Fig.2.8: Group photo

CHAPTER-3

BASICS OF SOLAR

3.1 LEARNING OUTCOMES

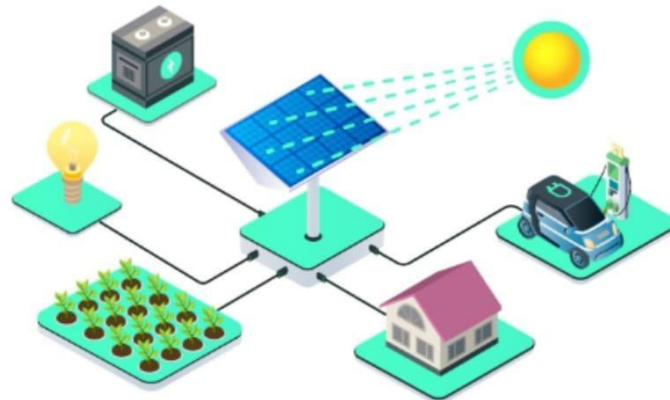


Fig.3.1: Solar energy system

1. Source of energy
2. Solar Energy
3. Basics of solar technologies and their application
4. How to estimate Solar capacity

3.2 HOW DOES ELECTRICITY REACH OUR HOMES

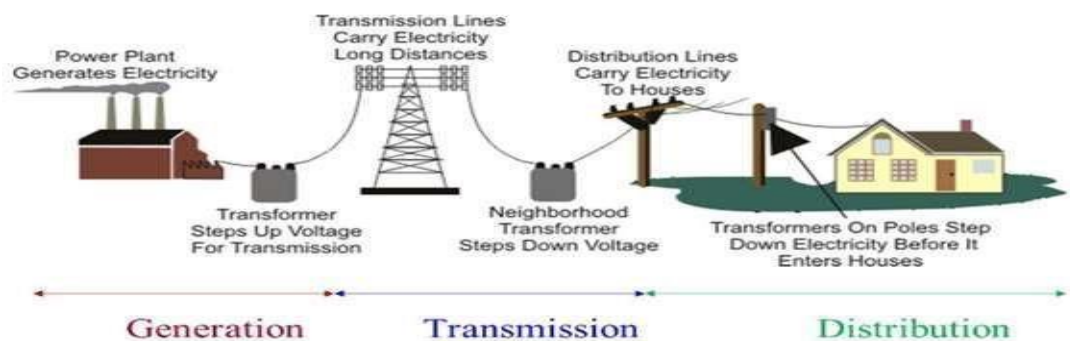


Fig.3.2: Diagram of Power system

Electricity reaches our homes through a complex system of generation, transmission, and distribution.

1. **Generation:** Electricity is generated at power plants using various sources of energy such as coal, natural gas, nuclear, hydroelectric, wind, and solar. In these plants, mechanical energy is converted into electrical energy through generators.
2. **Transmission:** Once electricity is generated, it is transmitted at high voltages over long distances through a network of transmission lines. High-voltage transmission lines are used to minimize energy loss during long-distance transportation. These lines can span hundreds or even thousands of miles, connecting power plants to substations.
3. **Substations:** At substations, the voltage of the electricity is reduced using transformers before being distributed to homes and businesses. Substations also help regulate and control the flow of electricity through the grid.
4. **Distribution:** After voltage is reduced at substations, electricity is distributed to homes and businesses through a network of distribution lines. These lines carry lower-voltage electricity and are typically installed on utility poles or buried underground.

Problems faced in generating electricity by coal



Fig.3.3

1. Limited coal supplies to last only next 70 years
2. Releases greenhouse gases like CO₂, CO etc. causing pollution and global warming eventually
3. Centralized generation
4. Infrastructure heavy

3.3 RENEWABLE ENERGY SOURCES



Fig.3.4: Renewable energy source

Renewable energy sources are energy resources that are naturally replenished on a human timescale, making them sustainable alternatives to finite fossil fuels.

Solar Energy: Solar energy harnesses sunlight to generate electricity through photovoltaic (PV) cells or solar thermal systems. PV cells convert sunlight directly into electricity, while solar thermal systems use sunlight to heat a fluid, producing steam to drive turbines and generate electricity.

Wind Energy: Wind energy utilizes the kinetic energy of wind to turn turbines and generate electricity. Wind turbines are commonly installed in wind farms, both onshore and offshore, to capture the energy from the wind.

Hydropower: Hydropower, or hydroelectric power, harnesses the energy of flowing water to generate electricity. It can be generated from large-scale dams and reservoirs, as well as smaller run-of-the-river or micro-hydro systems.

Advantages:

Environmental Benefits: Perhaps the most significant advantage of renewable energy sources is their minimal environmental impact. Unlike fossil fuels, which emit greenhouse gases and other pollutants when burned, renewable energy sources such as solar, wind, and hydropower produce little to no air or water pollution, helping to mitigate climate change and improve air quality.

Abundance and Sustainability: Renewable energy sources are derived from natural processes that are constantly replenished, making them inexhaustible and sustainable in the long term. Solar energy, wind energy, and hydropower are abundant and widely available, offering a virtually limitless supply of energy.

Energy Security: By diversifying energy sources and reducing reliance on imported fossil fuels, renewable energy sources enhance energy security and independence. They help mitigate the geopolitical risks associated with fossil fuel extraction, transportation, and supply disruptions.

Economic Benefits: The renewable energy sector creates jobs, stimulates economic growth, and drives innovation. Investments in renewable energy infrastructure and technologies generate employment opportunities in manufacturing, construction, installation, and maintenance, contributing to local economies and fostering technological advancement.

Cost-Effectiveness: The cost of renewable energy technologies, such as solar photovoltaics (PV) and wind turbines, has declined significantly in recent years, making them increasingly competitive with conventional fossil fuels. As technology advances and economies of scale are achieved, the cost of renewable energy is expected to continue decreasing, further improving its cost-effectiveness.

3.4 SOLAR ENERGY

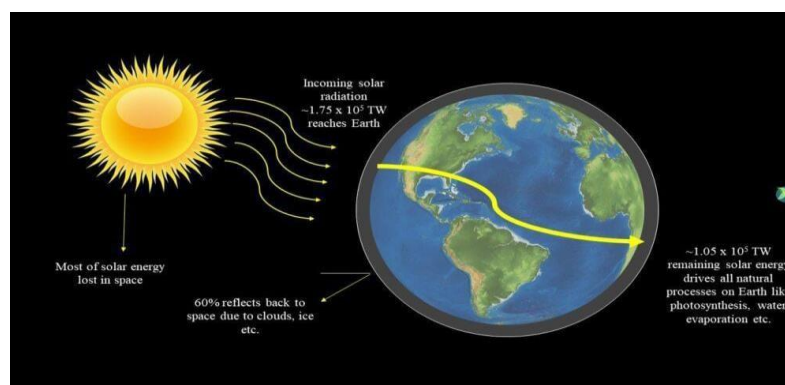


Fig.3.5

Solar energy is a renewable energy source derived from the sun's radiation. It is harnessed using various technologies to generate electricity, heat water, and provide space heating and cooling.

- Solar energy reaches Earth in two forms - heat and sunlight
- Solar PV uses the light energy from the Sun
- A hotter location may not necessarily be an ideal location for solar

SOLAR THERMAL



Fig.3.6: Solar thermal

Solar thermal energy is a technology that harnesses sunlight to generate heat, which can be used for various purposes including heating water for domestic or industrial use, space heating, and generating electricity through steam turbines.

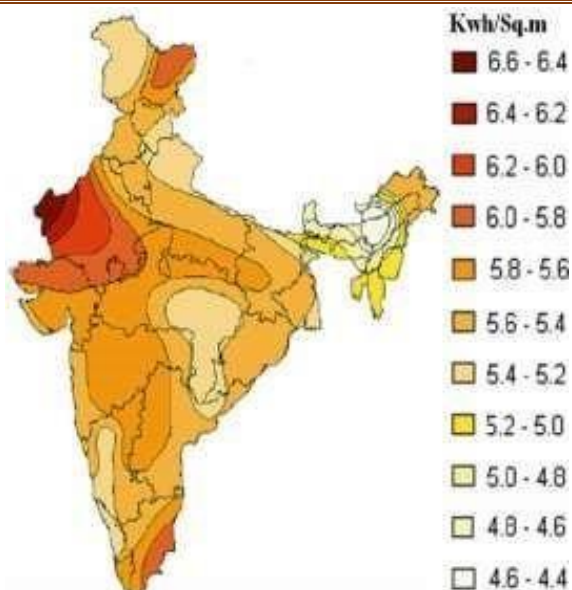
SOLAR PV(Photovoltaic)



Fig.3.7: Solar PV

Solar PV, or photovoltaic, is a technology that converts sunlight directly into electricity using solar panels.

- India receives abundant sunlight
Tropical Country
- About 5,000 trillion kWh per year energy is incident over India's land area
- Most parts receiving 4-7 kWh per sq. m per day.
- National Institute of Solar Energy has assessed the Country's solar potential of about 748 GW assuming 3% of the wasteland area to be covered by Solar PV modules.



3.5 BASIC TERMINOLOGIES

#	Particulars	Description	Unit
1	Current (I)	Flow of charged particles or electrical charge	ampere (A)
2	Voltage (V)	The amount of potential energy between two points on a circuit.	volt (V)
3	Power (P)	The rate at which energy is transferred $P = V \times I$, $P = E / t$	watt (W)
4	Electrical Energy (E)	Energy derived as a result of movement of electrons $E = q \times V$	joule (J) watt-hour (Wh) kilowatt-hour (kWh)

POWER	ENERGY
Power is the rate at which energy is transferred.	Energy is the ability to perform a work.
Power is measured in Watts or Kilowatts	Energy is measured in Watt Hours or Kilowatt hours
Example: My car's battery can provide 500 amps at 12 volts, which equals 6 kW of power.	Example: I left a 60W light Bulb on for 30 days, which raised my electricity bill by 43.2 kWh.

Fig.3.8: Basic Terminologies

3.6 PHOTOVOLTAIC EFFECT

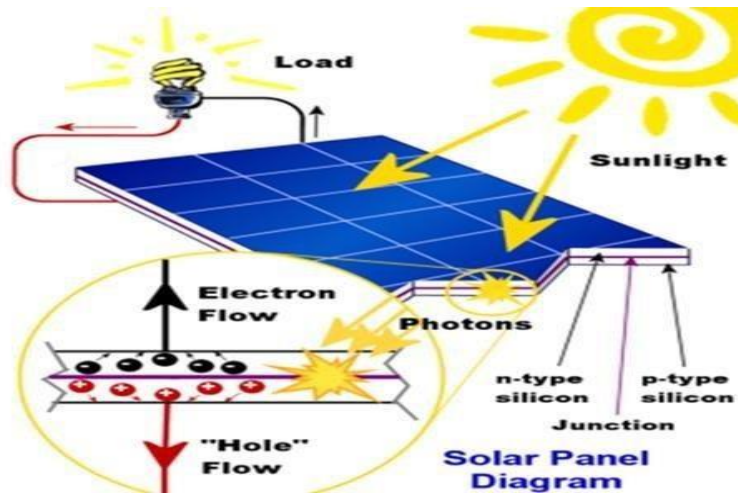


Fig.3.9:Photovoltaic Effect

The photovoltaic effect was first observed by French physicist Alexandre-Edmond Becquerel in 1839. The photovoltaic effect is the phenomenon by which certain materials produce an electric current when exposed to light. It is the underlying principle behind the functioning of solar photovoltaic (PV) cells, which are used in solar panels to convert sunlight directly into electricity.

3.7 SOLAR POWER PLANT CAPACITY

Case Study:

Nisha owns a house in Delhi with a terrace area of 500 Sq Ft and receives a monthly average bill of Rs. 6300. The sanctioned load for the house is 4kW. What is possible solar capacity required by Nisha to mitigate her consumption?



Fig.3.10: Power Estimation

As per thumb rule:

1 kWp needs 100 sq ft.

Here we have 500 Sq.ft, so can install 5kWp plant

(basis roof space available)

Every 1kWp generates 4 kWh/day (Approx 1400 – 1500 kWh per year)

Considering, the unit tariff is Rs. 7/kWh

Monthly bill: Rs. 6300

Monthly unit consumption: $6300/7 = 900$ units (kWh)

Daily unit consumption: $900/30 = 30$ units (kWh)

Solar capacity required: $30/4 = 7.5$ kWp

We can only install as per the lowest number, i.e. 4 kWp because Delhi DISCOM has a cap on the sanctioned load. In this case only 40% of the energy requirement will be fulfilled by solar.

Solar Power Plant Capacity: Step-by-step calculation

1. Roof area

- 1kWp needs ~ 100 Sq.ft shadow free area
- Capacity basis roof area = Roof area in Sq.ft/100

2. Electricity Consumption

- 1 kWp generates 4 kWh/day
- Monthly unit consumption = Monthly electricity bill amount/unit tariff
- Daily unit consumption = Monthly unit/30
- Capacity basis electricity consumption = Daily units/4

3. Sanctioned Load

- Maximum capacity should be as per the sanction load (as per state dependent netmetering regulation)

4.Final Capacity

- Final capacity = Lowest of the capacities identified in steps 1-3

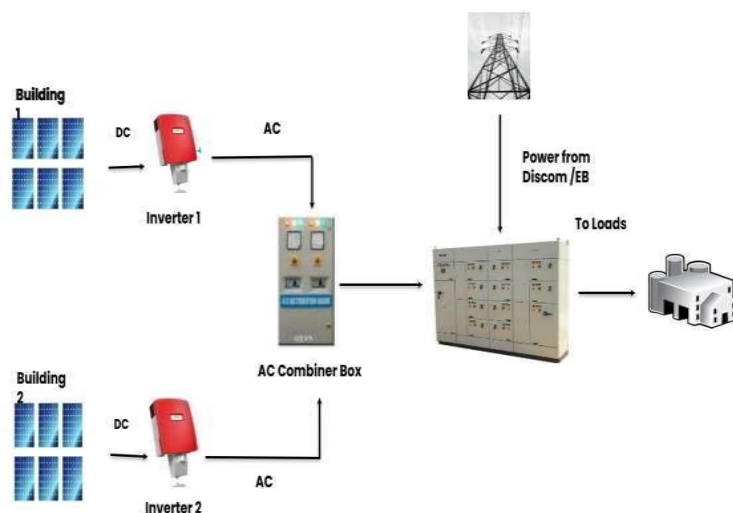


Fig.3.11: Solar power system

3.8 NET METERING ARRANGEMENT

Month	Nov
Self Consumption	500 units
Solar generation	600 units
Difference	+ 100 units
Energy bill	Rs. 0
Exported and carry forwarded to grid.	100 units

Month	Dec
Self Consumption	800 units
Solar generation	600 units
From previous month	100 units
Difference	-100 units
Energy bill	Rs. 100*5 =Rs.500



Fig.3.12: Net Meter

3.9 SOLAR PANELS / MODULES

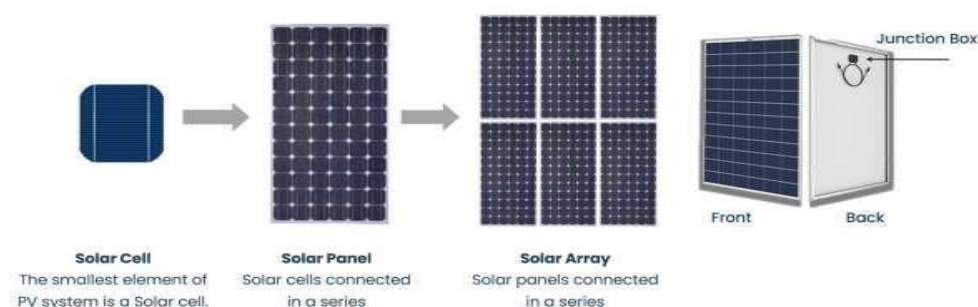


Fig3.13: Solar panel system

Types of crystalline modules

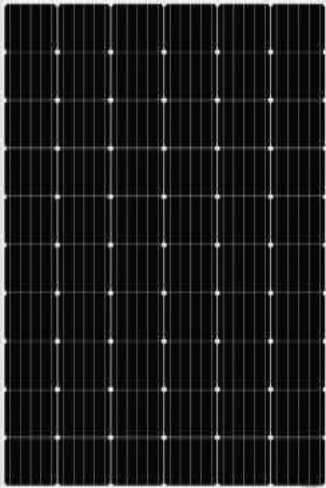
Sl.no.	Particulars	Polycrystalline	Monocrystalline
1	Picture		
2	Manufacturing	Made from multiple crystals	Made from single crystal ingot
3	Visual-Color	Blue	Black
4	Efficiency	Average	Slightly better than polycrystalline.
5	USP	-	Performs better in low light conditions.
6	Cost	Average	Higher than polycrystalline

Table 3.1: Types of crystalline modules

3.10 MODULE MOUNTING STRUCTURE (MMS)



Fig.3.14: Module Mounting Structure

Module mounting structures, also known as solar panel mounting systems or racking systems, are essential components of solar photovoltaic (PV) installations. They provide support and securement for solar panels, ensuring optimal positioning, stability, and durability.

Types of Mounting Systems:

1. **Roof Mounts:** Roof-mounted solar panel systems are installed directly onto the roof of a building. There are various types of roof mounts, including flush mounts, tilt mounts, and ballasted mounts, depending on the roof type, pitch, and installation requirements.
2. **Ground Mounts:** Ground-mounted solar panel systems are installed on the ground using structural supports such as poles, beams, or concrete foundations. Ground mounts are suitable for locations where rooftop space is limited or when specific orientations and tilts are required for optimal sunlight exposure.
3. **Tracking Systems:** Tracking systems automatically adjust the position of solar panels throughout the day to track the movement of the sun and maximize energy generation. There are two main types of tracking systems: single-axis trackers, which track the sun's movement along one axis (typically east-west), and dual-axis trackers, which track both horizontal and vertical movement.

Module mounting structures are critical components of solar PV systems, providing the foundation for securely mounting solar panels and optimizing energy generation. Proper design, installation, and maintenance of mounting structures are essential to ensure the long-term performance and reliability of solar PV installations.

3.11 INVERTER



Fig.3.15: Inverter

An inverter is an essential component of a solar photovoltaic (PV) system that converts the direct current (DC) electricity generated by solar panels into alternating current (AC) electricity suitable for use in homes, businesses, or the grid.

3.12 REVERSIBLE REDUCTIONS IN OUTPUT POWER

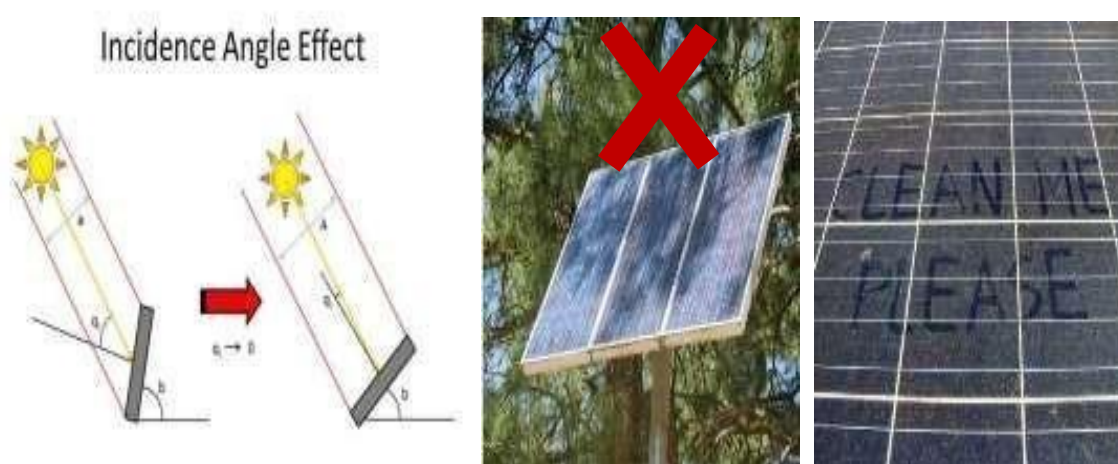


Fig.3.16: Reversible Reductions In Output Power

3.13 OVERVIEW OF FENICE ENERGY PRODUCTS

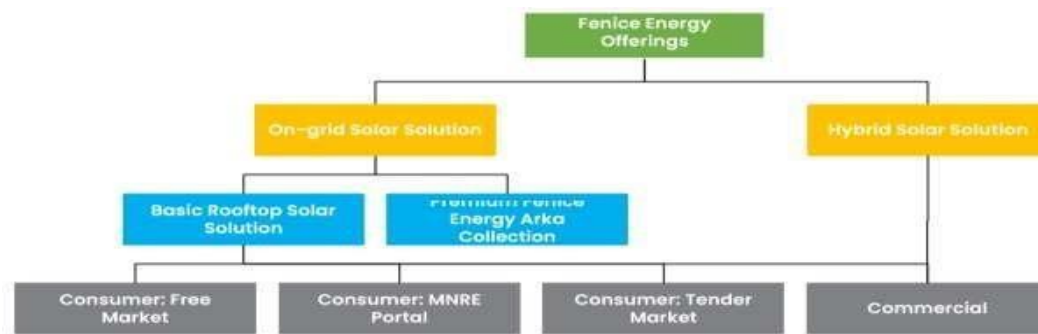


Fig.3.17: Overview of Fenice Energy products

3.14 SUBSIDY AND EMPANELMENT

Two Processes:

1. Tender-based process

- Existing process
- Vendor has to participate in tender called by the state discom to be able to sell systems with subsidy. Price is also capped at the lowest rate discovered
- Vendor collects the subsidy after plant commissioning and necessary approvals

Fenice Energy entities are empaneled under the tender process in Telangana, Kerala, New Delhi, Odisha, Mumbai and Goa .

2. National portal (MNRE)

- Newly launched on 31st July 2022
- Consumers can directly get subsidy by applying via national portal and get subsidy directly to their bank account
- Vendors just have to register in the portal by paying bank guarantee of INR 2.5 lacs at the DISCOM level

Fenice Energy is getting registered in the national portal across the country wherever Fenice Energy is present currently.

3.15 DIFFERENCE IN MODULE, PRICING AND PAYMENT PROCESS FOR FREE, MNRE AND TENDER MARKETS

	Free Market	MNRE Market (Via National portal)	Tender Market
Selling price	No restriction on selling price	No restriction on selling price	Can sell as per the rate discovered by the respective state tender
Fenice Energy price	As per Fenice Energy rate card	As per Fenice Energy rate card	Rate cards available with respective state sales teams
Subsidy amount	No subsidy	Fixed subsidy amount (INR per kW)	40% of discovered rate or benchmark prices for systems up to 3 kW and 20% for 3-10 kW
Subsidy payment and invoice amount	As per standard process	Customer is invoiced for 100% of the amount and subsidy is given to the customer directly by the government via national portal	Customer is invoiced for Total amount minus the subsidy. Subsidy is dispersed to the vendor after commissioning of plant.
Permitted modules	All domestically manufactured modules that are in the ALMM list (Approved list of module manufacturers) Non-DCR allowed	Only DCR modules Has to be in ALMM list	Only DCR modules Has to be in ALMM list

Table.3.2: Difference in module

DCR: Cell and Module manufactured in India

Non-DCR: Cell is imported and only module is manufactured in India

DCR - Domestic Content Requirement

ALMM: Only ALMM modules can be used for net metering projects since October 2022

3.16 PHOTOS



Fig.3.18: Photo 1



Fig.3.19: Photo 2



Fig.3.20: Photo 3



Fig.3.21: Photo 4

CHAPTER-4

MY ENTREPRENEUR JOURNEY

4.1 INTRODUCTION



Fig.4.1

In ‘My Entrepreneur Journey’, the speaker Sai Manoj C who is the founder of “Orenda Consulting and Contracting” and Co-founder of “Mod My Space Interiors Pvt. Ltd”. The speaker started session by narrating his college days, how it took 5 years to complete engineering which was of 4 years of period. After completing B.E in Mechanical engineering and not having any jobs in hand for around 6 to 8 months and started looking job at other domains and got placed in ‘Ipools which is a Swimming Pool construction company and worked for 2 years. After that resigned Ipools and started his own company called as ‘Orenda Consulting and Contracting’. In initial stages company faced some problems like getting clients and recruiting reliable and skilled labour. After sometimes company got some good clients and all was going well then the pandemic period arrived, during this company faced some huge losses due to lockdown and other safety protocols imposed by government. During this period speaker thought of doing higher studies in abroad, due not being financially stable started looking for scholarships and attend a contest called ‘Next Big game Changer in the Indian

Business Eco-system’. It was a all India competition conducted by Flinders University of South Australia in 2019. This prestigious Business World Scholarship was presented by Australian High Commission to India at New Delhi, and Mr. Sai Manoj was selected for a full scholarship of Rs. 35 Lakhs/-

During his MBA program, Mr. Sai Manoj interned at the New Venture Institute (NVI) of South Australia. After which went on to mentor several start-up founders in Australia to grow and expand their businesses.

This session made me understand that Entrepreneurship is not just about starting a business; it's a mindset, a way of life, and a journey filled with challenges and rewards. Today, we're going to explore what it means to be an entrepreneur and how you can cultivate the mindset and skills needed to succeed in this exciting journey.

It embrace that failure is not the opposite of success; it's a stepping stone to success. Embrace failure as an opportunity to learn, grow, and pivot. Remember, every setback is a lesson in disguise, and resilience is key to overcoming obstacles on the entrepreneurial journey.

4.2 OBJECTIVES

- 1.To enhance participants' entrepreneurial skills, such as creativity, critical thinking, problem solving, communication, leadership, and resilience.
- 2.To raise awareness of the challenges and opportunities associated with entrepreneurship, including market research, funding, legal considerations, and risk management.
- 3.To empower participants with the confidence, mindset, and resources they need to overcome obstacles, take calculated risks, and pursue their entrepreneurial goals with determination.
- 4.To emphasize the importance of ethical business practices, social responsibility, and sustainability in entrepreneurship, encouraging participants to create businesses that have a positive impact on society and the environment.



Fig.4.2

4.3 OUTCOMES

- 1.To enhance entrepreneurial skills, such as idea generation, market research, business planning, financial management, marketing, and networking.
- 2.To business ideas, validate assumptions, and gain insights into market demand, competition, and feasibility.
3. To develop an entrepreneurial mindset characterized by creativity, adaptability, problem solving, opportunity recognition, and a willingness to take calculated risks.
- 4.To build confidence in their ability to start and grow a business, overcome obstacles, and adapt to change, leading to increased self-efficacy and resilience.
- 5.Students were equipped with the capability to identify and address significant problems within their chosen domains and industries, thereby fostering a culture of innovation and solution-oriented thinking.



Fig.4.3

CHAPTER-5

5.1 CONCLUSION

The "Alcohol Detection with Motor Locking" system offers significant benefits to society by addressing the issue of drunk driving and reducing road accidents. By automatically preventing individuals under the influence of alcohol from starting their vehicles, this system contributes to enhanced road safety and a decrease in the number of accidents and fatalities. Additionally, it helps mitigate the loss of both lives and property, including damage to roads and vehicles. Furthermore, the implementation of such a system indirectly impacts the economy positively by reducing the costs associated with road accidents, such as medical expenses, repair costs, and lost productivity. Moreover, it prevents individuals from facing legal consequences and penalties resulting from drunk driving incidents, thereby averting potential life-altering consequences. Overall, the "Alcohol Detection and Engine Locking" system plays a vital role in promoting road safety, fostering economic growth, and preventing tragic outcomes associated with drunk driving incidents, making it a valuable asset for society.

The Auto EV India 2023 exhibition showcased the latest advancements and innovations in the electric vehicle industry, including new electric car models, battery technologies, charging infrastructure, and sustainable mobility solutions tailored for the Indian market. Exhibitions like these play a crucial role in promoting awareness, fostering collaboration, and driving the adoption of electric vehicles to combat pollution and reduce dependence on fossil fuels in India.

Solar energy offers a sustainable and renewable source of power that holds significant potential for addressing energy needs while mitigating environmental impact. Its widespread adoption has been driven by technological advancements, decreasing costs, and increasing efficiency of solar panels. Solar energy contributes to reducing greenhouse gas emissions, diversifying energy sources, and enhancing energy security. Additionally, solar power systems can be deployed at various scales, from small residential installations to large utility-scale projects. Despite its many benefits, challenges such as intermittency, energy storage, and initial investment costs remain. Continued research, innovation, and supportive policies are crucial for maximizing the benefits of solar energy and accelerating its integration into the global energy mix.

The resource persons provided insightful knowledge on "My Entrepreneurship Journey". This session typically cover various aspects of entrepreneurship, including ideation, business planning, market research, funding options, marketing strategies, and leadership development. Through interactive discussions, workshops, mentorship, and networking opportunities, participants can learn from experienced entrepreneurs, industry experts, and peers, while also receiving feedback on their ideas and business plans.

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